Digital Table: Saving Data Forever

PYTHON & CSV | SESSION 1

Welcome, Data Detective! [cite: 859] Have you ever noticed that when you restart your Python program, your variables "forget" everything? [cite: 853] Variables are like chalk on a blackboard—they are wiped clean easily! [cite: 853]

The Memory Problem

Variables live in "Temporary Memory." [cite: 837] When the computer stops, your data disappears like magic! [cite: 854] But what if you wanted to save your score or a list of your friends' birthdays forever? [cite: 857]

> **The Solution: Permanent Storage!**
> Writing data to a file is like writing in a notebook. Even after the program ends, the "notebook" stays safe. [cite: 857] Today, we use **CSV files** to remember everything! [cite: 858]

[Image of chalkboard vs notebook analogy]

What is a CSV File?

CSV stands for **"Comma Separated Values"**. [cite: 859] It is the most common way for computers to share table-like data. [cite: 836]

THE DIGITAL TIFFIN BOX

Think of a CSV file as a digital tiffin box! [cite: 860] Just like your tiffin has compartments separated by walls, a CSV file has data separated by **commas**. [cite: 861]

> **Example:**
> `Rahul, 10, Mumbai` [cite: 863]
>
> Each comma creates a new compartment for your information! [cite: 862]

Rows and Columns

CSV files look just like your school timetable! [cite: 865, 866]

- **Rows:** Horizontal lines going across (like each day). [cite: 865]
- **Columns:** Vertical sections going down (like each period). [cite: 866]

[Image of school timetable structure]

Seeing Inside a CSV File

A CSV file is really just a text file. If you open it in a simple app like Notepad, it looks like this: [cite: 895, 896]

```
Name, Marks, Grade [cite: 897]
Aarav, 85, A [cite: 898]
Priya, 72, B [cite: 899]
Rohan, 91, A [cite: 900]
```

But when you open that same file in Excel or Google Sheets, the computer turns those commas into neat table cells! [cite: 903, 904]

INDIAN DATASET EXAMPLES

- **Attendance Register:** Rows for students and columns for Roll Number and Presence. [cite: 908, 909]
- **IPL Points Table:** Team Name, Matches Won, and Points—just like on TV! [cite: 910, 911]
- **Bazaar List:** Item Name, Quantity, and Price. [cite: 912, 913]

Reading CSV Files in Python

Python has a built-in helper called the **'csv' module** to read these files easily. [cite: 915, 916]

THE "SAFE OPEN" RULE

We use the `with` open command to open our files. This is like opening a box safely—it makes sure the box is closed properly when we are done! [cite: 917]

```
import csv [cite: 919]

with open('marks.csv') as file: [cite: 920]
    reader = csv.reader(file) [cite: 921]
    for row in reader: [cite: 922]
        print(row) [cite: 923]
```

This code will print every row of your table as a Python List! [cite: 929]

Picking Your Columns

When Python reads a row, it treats it like a list. Each column has an **Index Number** starting from **0**. [cite: 929, 930]

EXAMPLE ROW:

`['Aarav', '85', 'Pass']` [cite: 933]

- `row[0]` is the first column (Aarav). [cite: 927, 934]
- `row[1]` is the second column (85). [cite: 928, 935]

CODE-ALONG: READING STUDENT MARKS

```
# Let's print only names and marks!
with open('student_marks.csv') as file: [cite: 941]
    reader = csv.reader(file) [cite: 942]
    for row in reader: [cite: 943]
        print(row[0], row[1]) [cite: 944]
```

Output: Aarav 85, Priya 92, Rohan 78, Ananya 88 [cite: 945]

The Data Detective Challenge

Now it's your turn to write some logic! Can you help the teacher mark the report card? [cite: 946]

YOUR TASK:

Open `student_marks.csv` and loop through the data. [cite: 946]

- If marks are **greater than 40**, print "Pass". [cite: 946]
- If marks are **40 or less**, print "Fail". [cite: 946]

DATASET SNIPPET:

Name	Marks
Ishaan	85
Ananya	38
Kabir	72

Hint: Use `int(row[1])` *to convert the text to a number before comparing!*

What Did We Learn Today?

- A **CSV** file is a blueprint for storing tables in text form. [cite: 836]
- Variables are temporary, but **Files** are permanent. [cite: 837, 857]
- Python's **csv module** makes reading tables easy. [cite: 915, 916]
- We can pick specific data using **Index Numbers**. [cite: 929]

WHY CSV?

Programmers love CSV files because they are super helpful for saving and sharing data between different apps! [cite: 947, 948]

Shabaash! You are now a Data Expert! [cite: 949]

COMING SOON: SESSION 2

Next up, we will learn how to **Write** our own data to a CSV file! [cite: 950]